PROBLEM FS7

Wedge between Two Liquids

Original statement · fluid_statics_7.pdf

Inclined surfaces | Two liquids | Force and moment | Virtual column

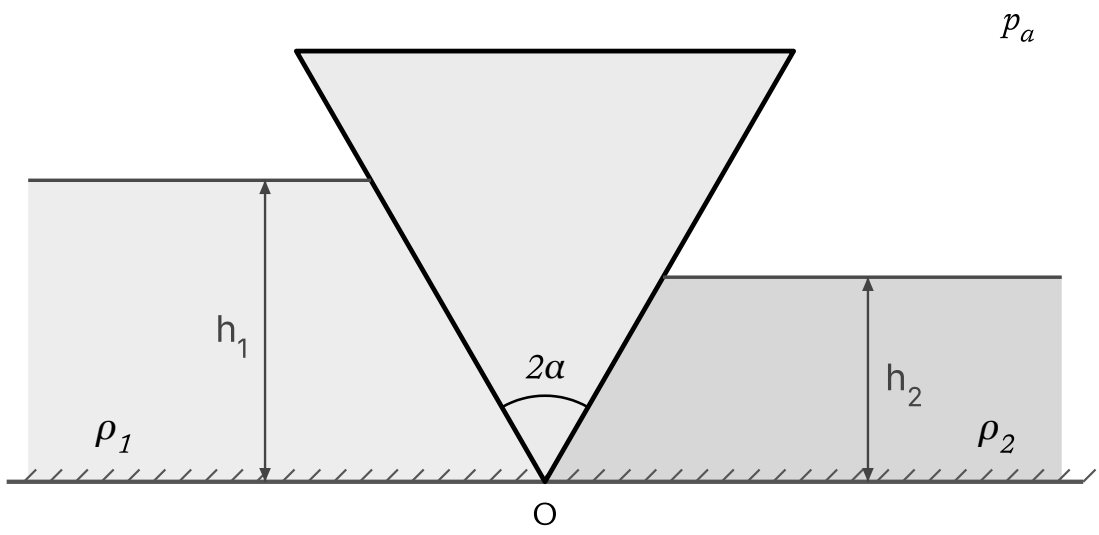
OFFICIAL PROBLEM STATEMENT

A two-dimensional wedge of angle 2α and weight W per unit length rests on its lower vertex in the position shown in the figure. The wedge separates two liquids of densities ρ_1 and $\rho_2 > \rho_1$. The levels of the two liquids are h_1 and h_2 respectively. Find:

1. The reaction force exerted on the wedge at the support point O.

2. The relation that must hold between h_1 and h_2 for the equilibrium position shown in the figure to be possible.

Translated from the Spanish original.



Statement figure (redrawn from fluid_statics_7.pdf, p. 1): wedge of angle 2α and weight W per unit length resting on its vertex O; liquid ρ_1 of depth h_1 on the left and $\rho_2 > \rho_1$ of depth h_2 on the right

Source: fluid_statics_7.pdf (pages 1-2, sheet of 05-09-07, problem P5). Notes.pdf, Chapter 5: Eqs. 5.10, 5.11 and 5.14.

Worked solution

Hypotheses, Data and Unknowns

- Both liquids at rest, incompressible; air at p_a above them and on the top face of the wedge, so the p_a contributions cancel over the closed surface and only gauge pressures count.
- Planar problem: forces and moments per unit length normal to the paper.
- The wedge is symmetric about the vertical through O, each face making the angle α with that vertical; its centre of gravity lies on that vertical, so W has no moment about O.
- At O the wedge rests on the floor; the reaction $\vec{R} = R_x\vec{e}_x + R_y\vec{e}_y$ acts at O (no moment at a point support).
- Data: α, W, ρ_1, ρ_2, h_1, h_2, g. Unknowns: R_x, R_y and the relation between h_1 and h_2.

Symbol	Meaning	SI unit
α	half-angle of the wedge	rad
W	weight per unit length	N/m
h_1, h_2	liquid levels above O	m
R_x, R_y	reaction at O per unit length	N/m

Coordinates, Element and Normal of Each Face

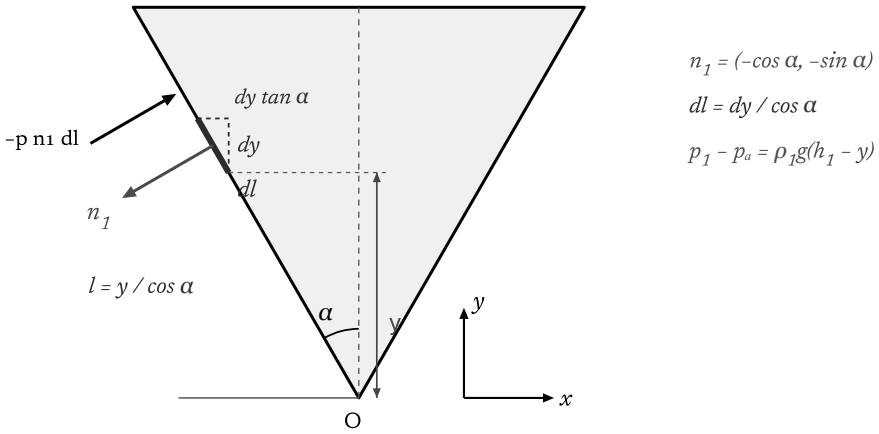
Origin at the vertex O, x horizontal to the right, y vertical and up. A point of the left face at the height y is at $x = -y\tan\alpha$: in the right triangle with the vertical leg y (adjacent to α) and the horizontal leg (opposite), the horizontal leg is $y\tan\alpha$, and the hypotenuse, the distance from O along the face, is $l = y/\cos\alpha$.

The element of the face between y and $y + dy$ is the hypotenuse of the small right triangle with the legs dy and $dy\tan\alpha$, so $dl = dy/\cos\alpha$.

The face makes the angle α with the vertical, so its unit normal makes the angle α with the horizontal. Liquid 1 is on the left and below the face, so the normal towards liquid 1 is

$$\vec{n}_1 = -\cos\alpha\,\vec{e}_x - \sin\alpha\,\vec{e}_y, \qquad \vec{n}_2 = \cos\alpha\,\vec{e}_x - \sin\alpha\,\vec{e}_y$$

($\vec{n}_2$ by mirror symmetry for the right face). Check: $\vec{n}_1$ is perpendicular to the face direction $(-\sin\alpha, \cos\alpha)$, since $\sin\alpha\cos\alpha - \sin\alpha\cos\alpha = 0$.



Left face of the wedge, making the angle α with the vertical. The element dl is the hypotenuse of the right triangle with the vertical leg dy (adjacent to α) and the horizontal leg $dy\tan\alpha$ (opposite), so $dl = dy/\cos\alpha$; for the same reason the distance from O along the face is $l = y/\cos\alpha$. The unit normal towards liquid 1 is $\vec{n}_1 = -\cos\alpha\,\vec{e}_x - \sin\alpha\,\vec{e}_y$ and the force on the wedge is $-p\vec{n}_1\,dl$

By (5.10), the gauge pressures are $p_1 - p_a = \rho_1 g(h_1 - y)$ for $0 \le y \le h_1$ and $p_2 - p_a = \rho_2 g(h_2 - y)$ for $0 \le y \le h_2$.

Forces on the Faces

On the left face, $d\vec{F}_1 = -(p_1 - p_a)\vec{n}_1\,dl = \rho_1 g(h_1 - y)(\cos\alpha\,\vec{e}_x + \sin\alpha\,\vec{e}_y)\,\dfrac{dy}{\cos\alpha}$, so

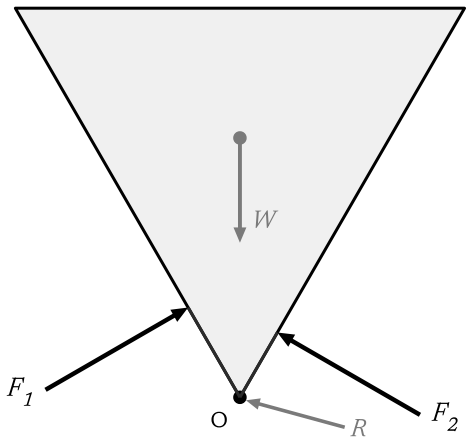
$$\vec{F}_1 = \rho_1 g\,(\vec{e}_x + \tan\alpha\,\vec{e}_y)\int_0^{h_1}(h_1 - y)\,dy = \rho_1 g\,(\vec{e}_x + \tan\alpha\,\vec{e}_y)\left[h_1 y - \frac{y^2}{2}\right]_0^{h_1} = \frac{\rho_1 g h_1^2}{2}\,(\vec{e}_x + \tan\alpha\,\vec{e}_y)$$

and in the same way on the right face, with $-\vec{n}_2 = -\cos\alpha\,\vec{e}_x + \sin\alpha\,\vec{e}_y$:

$$\vec{F}_2 = \frac{\rho_2 g h_2^2}{2}\,(-\vec{e}_x + \tan\alpha\,\vec{e}_y)$$

Check with the projection rule (5.14). The horizontal component of $\vec{F}_1$ is the force on the vertical projection of the face, a rectangle of height h_1 and unit width: $\rho_1 g h_1^2/2$. The vertical component is upward and equals the weight of the **virtual** column of liquid 1 above the face, the triangle with the vertical side h_1 and the horizontal side $h_1 \tan\alpha$: $\rho_1 g \cdot \frac{1}{2} h_1 \cdot h_1 \tan\alpha$. Both agree.

1. The Reaction at O



Free-body diagram of the wedge (per unit length): F_1 and F_2 normal to the faces, applied at the centres of pressure, at the distances $h_1/(3\cos\alpha)$ and $h_2/(3\cos\alpha)$ from O along the faces; the weight W on the symmetry line through O; the reaction R at O

Force balance of the wedge, $\vec{F}_1 + \vec{F}_2 - W\vec{e}_y + \vec{R} = 0$, component by component:

$$R_x = -\frac{\rho_1 g h_1^2}{2} + \frac{\rho_2 g h_2^2}{2}, \qquad R_y = W - \frac{\rho_1 g h_1^2}{2}\tan\alpha - \frac{\rho_2 g h_2^2}{2}\tan\alpha$$

2. The Moment about O and the Condition on the Levels

Every $d\vec{F}_1$ is normal to the face, so its arm about O is the distance along the face, $l = y/\cos\alpha$, and it turns the wedge clockwise (towards liquid 2). Its moment is

$$M_1 = \int_0^{h_1} (p_1 - p_a)\, l\, dl = \int_0^{h_1} \rho_1 g (h_1 - y)\, \frac{y}{\cos\alpha}\, \frac{dy}{\cos\alpha} = \frac{\rho_1 g}{\cos^2\alpha} \left[\frac{h_1 y^2}{2} - \frac{y^3}{3} \right]_0^{h_1} = \frac{\rho_1 g h_1^3}{6\cos^2\alpha}$$

The same follows from the cross product $\vec{r} \times d\vec{F}_1$ with $\vec{r} = (-y\tan\alpha, y)$: $M_z = x\, dF_y - y\, dF_x = -\rho_1 g (h_1 - y)\, y\, (\sin^2\alpha + \cos^2\alpha)\, dy/\cos^2\alpha$, clockwise, with $\sin^2\alpha + \cos^2\alpha = 1$. By symmetry, liquid 2 gives the counterclockwise moment $M_2 = \rho_2 g h_2^3/(6\cos^2\alpha)$. The weight and the reaction pass through O. Moment balance about O:

$$\frac{\rho_1 g h_1^3}{6\cos^2\alpha} = \frac{\rho_2 g h_2^3}{6\cos^2\alpha} \quad\Longrightarrow\quad \rho_1 h_1^3 = \rho_2 h_2^3 \quad\Longrightarrow\quad \frac{h_1}{h_2} = \left(\frac{\rho_2}{\rho_1}\right)^{1/3}$$

Answers and Checks

- $R_x = \frac{g}{2}(\rho_2 h_2^2 - \rho_1 h_1^2)$, $R_y = W - \frac{g}{2}(\rho_1 h_1^2 + \rho_2 h_2^2)\tan\alpha$.
- $h_1/h_2 = (\rho_2/\rho_1)^{1/3}$: the lighter liquid must stand higher ($h_1 > h_2$, as drawn). The angle α does not appear.
- **Support conditions (extension, not asked in the key)**: a vertex resting on the floor can only push up, so the position is possible only if $R_y \geq 0$, that is $W \geq \frac{g}{2}(\rho_1 h_1^2 + \rho_2 h_2^2)\tan\alpha$, and only if the support can supply the horizontal force R_x (friction or a notch at O).
- With the levels in that relation, $\rho_2 h_2^2 = \rho_1 h_1^2\,(h_1/h_2) > \rho_1 h_1^2$, so $R_x > 0$: the net push of the liquids is towards liquid 1 and the floor holds the vertex back.
- **Limiting case** $\rho_1 = \rho_2$: $h_1 = h_2$, $R_x = 0$ and $R_y = W - \rho g h^2 \tan\alpha$, the weight minus the buoyancy of the submerged triangle of area $h^2\tan\alpha$ (Archimedes' principle, with the vertex as the only support).
- **Dimensions**: $\rho g h^2$ is (kg m^{-3})(m s^{-2})(m^2) = N/m, like W; $\rho g h^3$ is N m/m.

- **Check against the official key**: fluid_statics_7.pdf gives $\vec{F}_1 = \rho_1 g h_1^2 (\vec{e}_x + \tan\alpha\, \vec{e}_y)/2$, $\vec{F}_2 = \rho_2 g h_2^2 (-\vec{e}_x + \tan\alpha\, \vec{e}_y)/2$, the same R_x and R_y, $M_1 = \rho_1 g h_1^3/(6\cos^2\alpha)$ and $h_1/h_2 = (\rho_2/\rho_1)^{1/3}$, the same results. The key writes the normals pointing into the wedge, the opposite of the convention of Eq. 5.11; the forces are the same.